

Edward Tufte's Six Principles of Graphical Excellence and Cognitive Load Theory

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Data visualization is important towards how we understand information. Poorly designed visuals can confuse viewers, while well-designed visualizations can reveal trends and insights instantly. Edward Tufte, is a pioneer in information design, made six principles of graphical excellence that can be used as a guide in order to make effective data visualizations. These principles align closely with the Cognitive Load Theory that was coined in 1988 by John Sweller, which explains how human working memory processes information. This work discusses Tufte's principles and its examples as well as the cognitive load theory.

Edward Tufte's Six Principles of Graphical Excellence are:

1. **Show the Data**

The first principle emphasizes on presenting the data clearly and accurately. Tufte suggests that the visualizations should focus on the data itself rather than the decorative or stylistic elements that could obscure the data.

Example:

A simple line graph showing population growth over time is more effective than a heavily stylized chart with backgrounds, shadows or unnecessary designs.

2. **Maximize Data-Ink Ratio**

Tufte's second principle introduces the idea of data-ink ratio which refers to the proportion of "ink" or pixels that is used to represent actual data compared to the total ink used in the graphic. A high data-ink ratio means most visual elements directly represent data.

Example:

Removing excessive gridlines or thick borders from a bar chart increases the emphasis of the bars that show the actual data.

3. **Erase Non-Data Ink**

Following with the second principle, non-data ink includes decorative elements, heavy borders, backgrounds or redundant labels that do not convey information. These should be removed unless they serve a clear purpose

Example:

Eliminating 3D effects and background images in pie charts helps viewers interpret proportions more accurately.

4. **Multifunctioning Graphical Elements**

Tufte's fourth principle says that graphical elements should serve multiple purposes whenever possible, such as labeling, scaling and grouping data simultaneously to improve compactness.

Example:

Data points labeled directly with values and categories eliminate the need for separate legends and reduce eye movement.

5. Data Density and Small Multiples

Data density refers to presenting large amounts of information in a compact and organized manner. Small multiples are a series of similar graphs that allow easy comparison across datasets.

Example:

A grid of small line charts showing monthly sales trends for different regions allows quick comparison without overwhelming the viewer.

6. Use Clear, Detailed Labels

Labels should be clear, precise, and placed directly on the graphic whenever possible. Good labeling reduces ambiguity and improves comprehension.

Example:

Labeling data points directly on a chart is more effective than using a legend that requires constant cross-referencing.

Cognitive Load Theory is a model developed by John Sweller to describe how the human mind processes information. The term cognitive load refers to how much information can a working memory of a human process at any given moment. CLT states that there are three main parts to a memory such as the sensory memory, working memory, and long-term memory. There are also three types of cognitive load which are intrinsic, extraneous, and germane.

Main parts of the CLT:

1. **Sensory Memory** - refers to the memory of the things that happen around us that are often forgotten and only passes select information to the working memory.
2. **Working Memory** - the memory that can only store a limited amount of information for a short amount of time to be processed be moved to the long term memory or be forgotten.
3. **Long-term Memory** - type of memory that stores the information as structured schemas to organized information to be recalled later when needed.

Three Types of Cognitive Load:

1. **Intrinsic Load** - refers to the complexity of the information based on the number of variables or element it has.

Example: Learning how to drive car which requires knowledge in how to maneuver, traffic rules, operating the transmission, and seeing the traffic conditions all at the same time.

2. **Extraneous Load** - refers to the mental effort required to extract valuable information from unnecessary ones.

Example: Trying to read a graph with poorly design graphs and confusing colors.

3. **Germane Load**- refers to the mental effort required to connect newly gathered information from existing knowledge or schemas in long-term memory.

Example: Learning how to use a specific library in Python and integrate it in the project.

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